

Maike L Morrison

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Education

2020–2025 (expected)	Stanford University <i>PhD in Biology (Ecology & Evolution)</i> Dr. Noah Rosenberg, advisor	Stanford, CA
2016–2020	The University of Texas at Austin <i>BS in Mathematics, Dean's Scholars Honors Program</i> Drs. Lauren Ancel Meyers & Mark Kirkpatrick, advisors	Austin, TX

Research Experience

Sep 2020– Present	Graduate Research Assistant <i>Dr. Noah Rosenberg, Stanford University Department of Biology</i> Developing & applying variability statistics for ecology, population genetics, & microbiomes.	Stanford, CA
Mar 2023– Sep 2023	Visiting Scientist, International Agency for Research on Cancer <i>Dr. Nicolas Alcala, International Agency for Research on Cancer</i> Identified overlooked genomic signatures of carcinogens using novel statistical methods.	Lyon, France
Jun 2022– July 2022	Complex Systems Summer School Student <i>Santa Fe Institute</i> Explored the effects of spatial immune structure on infectious disease dynamics.	Santa Fe, NM
May 2019– Jul 2019	UCSF Amgen Scholars Program <i>Dr. John Witte, UC San Francisco Department of Biostatistics & Epidemiology</i> Conducted genome-wide association studies of immune response strength to 22 antigens.	San Francisco, CA
Jun 2018– Aug 2018	The Mathematical Biosciences Institute NSF REU <i>Dr. Ephraim Hanks, Penn State University Department of Statistics</i> Explored dynamics of disease spread over patchy terrain through simulation study.	State College, PA

Peer-Reviewed Publications

6. **Morrison ML**, Xue KS, Rosenberg NA (2025). Quantifying compositional variability in microbial communities with FAVA. *Proceedings of the National Academy of Sciences*, 122(11), e2413211122.
5. **Morrison ML** & Rosenberg NA (2023). Mathematical bounds on Shannon entropy given the abundance of the i th most abundant taxon. *Journal of Mathematical Biology*, 87(5), 76.
4. **Morrison ML**, Alcala N, Rosenberg NA (2022). FSTRUCT: an F_{ST} -based tool for measuring ancestry variation in inference of population structure. *Molecular Ecology Resources*, 22(7), 2614-2626.
3. Ingle T, **Morrison ML**, Wang X, Mercer T, Karman V, Fox S, Meyers LA (2021). Projecting COVID-19 isolation bed requirements for people experiencing homelessness. *PLOS One*, 16(5), e0251153.
2. Kachuri L*, Francis SS*, **Morrison ML**, Wendt GA, Bossé Y, Cavazos TB, Rashkin SR, Ziv E, Witte JS (2020). The landscape of host genetic factors involved in immune response to common viral infections. *Genome Medicine* 12(93), 1-18.
1. **Morrison M***, Castro LA*, Meyers LA (2020). Conscientious vaccination exemptions in kindergarten to eighth-grade children across Texas schools from 2012 to 2018: A regression analysis. *PLOS Medicine* 17(3), e1003049.

Preprints

3. **Morrison ML**, Mangé L, Senkin S, Rosenberg NA, Foll M, Fernandez-Cuesta L, Alcala N (2023). Variability of mutational signatures is a footprint of carcinogens. *medRxiv*.
2. Xue KS, Walton SJ, Goldman DA, **Morrison ML**, Verster AJ, Parrott AB, Yu FB, Neff NF, Rosenberg NA, Ross BD, Petrov DA, Huang KC, Good BH, Relman DA (2023). Prolonged delays in human microbiota transmission after a controlled antibiotic perturbation. *bioRxiv*.
1. Miyagi M, **Morrison M**, Kirkpatrick M (2022). How many ecological niches are defined by the super-abundant marine microbe *Prochlorococcus*? *bioRxiv*.

*These authors contributed equally

Teaching

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| Winter
2025 | BIO 192/387: Statistical Methods in Ecology & Evolution
<i>Instructor of Record</i> | Stanford University |
| | · Designed and taught course on the theory and application of biological diversity statistics. | |
| Fall
2023 | BIO 388: CEHG Seminar - Microbiome Ecology & Evolution
<i>Instructor of Record</i> | Stanford Univ.
<i>with Dr. Noah Rosenberg</i> |
| | · Co-organized Fall 2023 Computational, Evolutionary, & Human Genomics seminar. | |
| | · Facilitated accompanying class, including lecturing on statistical methods for biology. | |
| Winter
2022 | BIO 187: Mathematical Population Biology
<i>Graduate Teaching Assistant</i> | Stanford University
<i>for Dr. Noah Rosenberg</i> |
| | · Prepared and presented lectures on mathematical bounds on F_{ST} and Shannon entropy. | |
| Winter
2021 | BIO 85: Evolution
<i>Graduate Teaching Assistant</i> | Stanford University
<i>for Drs. Dmitri Petrov & Molly Schumer</i> |
| | · Wrote & graded assessments and organized discussion sections for introductory online course. | |

Awards & Fellowships

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| 2022 | Stanford Biology Department Excellence in Teaching Award (\$1K) |
| 2021–2025 | NSF Graduate Research Fellowship Program (\$138K) |
| 2020–2025 | Stanford Graduate Fellowship (\$250K) |
| 2020 | UT Austin College of Natural Sciences Distinctions in Service & Leadership, and in Research |
| 2020 | George H. Mitchell Award for Outstanding Undergraduate Research (\$5K) |
| 2020 | NSF Graduate Research Fellowship Program, Honorable Mention |
| 2019–2020 | Phi Beta Kappa, junior year election |
| 2019 | Barry M. Goldwater Scholarship (\$7.5K) |

R Packages

FAVA: An F_{ST} -based Assessment of Variability over relative Abundances

sigvar: Quantify and visualize variability of mutational signatures within and across samples

FSTruct: Measure variability in population structure estimates

Academic Service

2024–2025	Member, CEHG Symposium Organizing Committee <i>Center for Computational, Evolutionary, and Human Genomics (CEHG)</i>	Stanford University
2023–2024	Student Member, Ecology & Evolution PhD Admissions Committee	Stanford University
2021–2022	Co-Organizer, Eco-Evo Lunch	Stanford University
2021	Co-Organizer, Biology Pre-Orientation Program (BioPOP)	Stanford University

Peer review activity: *Theoretical Population Biology, Genetics*

Conference Presentations**Selected Oral Presentations**

May 2024	Modeling & Theory in Pop. Bio. (Banff Int. Research Station) “Quantifying the stability of microbiomes and the timescale of antibiotic perturbation” <i>Award for best talk by graduate student or postdoc</i>	Banff, AB
May 2024	American Mathematical Society Spring West. Sect. Meeting “Quantifying compositional variability in microbial communities” <i>Invited talk</i>	San Francisco, CA
Aug 2023	Ecological Society of America Annual Meeting “Quantifying compositional variability in microbial communities”	Portland, OR

Selected Poster Presentations

Mar 2024	Society for Molecular Biology & Evolution “Quantifying the stability of microbiomes and the timescale of antibiotic perturbation with FAVA”	Puerto Vallarta, Mexico
Mar 2024	The Allied Genetics Conference “Quantifying compositional variability in microbial communities”	Washington, DC
Dec 2022	The Human Microbiome: Ecology and Evolution (Keystone Symposia) “A diversity statistic for quantifying microbiome stability”	Banff, AB
June 2022	Population, Evolutionary, & Quantitative Genetics Conference “FSTruct: An F_{ST} -based tool for quantifying ancestry variability” <i>Genetics Society of America Poster Award</i>	Pacific Grove, CA
Jun 2019	Evolution Meeting “How many ecotypes are comprised by the world’s most abundant photosynthetic organism?” <i>Undergraduate Diversity at Evolution Award Program</i>	Providence, RI